

RICAM SYRUP 5 mg / 5 ml

VIRIC14-var1 (MY)

DESCRIPTION

Clear colourless syrup with Tutti-Frutti flavour.

COMPOSITION

Cetirizine hydrochloride 5 mg / 5 mL.

PHARMACODYNAMICS

Cetirizine hydrochloride, a human metabolite of hydroxyzine, is a potent antihistamine that selectively inhibits the effect of peripheral H₁-receptors.

Cetirizine belongs to the piperazine class of antihistamines and is the major carboxylated metabolite of hydroxyzine. However, it is less lipophilic than hydroxyzine and thus less likely to distribute into the central nervous system (CNS) and produce sedation.

Generally, antihistamines contain 1 or 2 heterocyclic or aromatic rings attached to the side chain of histamine. Cetirizine has more polar or less lipophilic functional groups attached to the ethylamine side chain which has resulted in less CNS penetration, greater selectivity for the H₁-receptor, and less anticholinergic side effects.

Cetirizine may also have a modulatory effect on inflammatory cells by inhibiting eosinophil migration or eosinophilotactic mediators. Additionally, cetirizine may also have inhibitory effects on other infiltrating inflammatory cells, such as neutrophils.

Cetirizine has shown to be safe for allergic patients with mild to moderate asthma and does not cause prolongation of QT interval.

PHARMACOKINETICS

Absorption

Cetirizine is rapidly absorbed from the gastrointestinal tract after oral doses and peak plasma concentrations occur within about an hour. Food delays the time to peak plasma concentrations but does not decrease the amount of drug absorbed.

Bioavailability

Oral: 70%.

The extent of bioavailability is similar when cetirizine is given as solutions, capsules or tablets.

Distribution

Cetirizine is highly bound to plasma proteins (93%). Volume of distribution is 0.5 to 0.8 L/kg.

It does not appear to cross the blood-brain barrier to a significant extent. It has been detected in breast milk.

Metabolism

Cetirizine is not extensively metabolized by the liver. The inactive O-dealkylated metabolite has been identified in both the plasma and feces, but not in the urine.

Elimination

Cetirizine is excreted primarily in the urine (60%) mainly as unchanged drug and has an elimination half-life of about 10 hours. Cetirizine is poorly cleared by hemodialysis.

Special populations

Children, infants and toddlers: The half-life of cetirizine was about 6 hours in children of 6-12 years and 5 hours in children 2-6 years. In infants and toddlers aged 6 to 24 months, it is reduced to 3.1 hours.

Renal insufficiency

The elimination half-life of cetirizine may be prolonged. Dosing adjustment is necessary in patients with moderate or severe renal impairment. Hemodialysis is unlikely to be of value in removing cetirizine from the plasma.

Elderly or hepatic insufficiency

The elimination half-life is prolonged by 50% in geriatrics and in patients with chronic liver disease as compared to normal healthy adults. Dosing adjustment is only necessary in hepatically impaired patients if concomitant renal impairment is present.

INDICATIONS

Adults and children of 2 years and above: symptomatic treatment of seasonal allergic rhinitis, perennial allergic rhinitis and urticaria of allergic origin.

CONTRAINDICATIONS

- History of hypersensitivity to any of the constituents of the formulation, to hydroxyzine or to any piperazine derivatives.
- Patients with severe renal impairment at less than 10 ml/min creatinine clearance.

WARNINGS AND PRECAUTIONS

- Activities Requiring Mental Alertness: In clinical trials the occurrence of somnolence has been reported in some patients taking Cetirizine: due caution should therefore be exercised when driving a car or operating potentially dangerous machinery.
- Caution in epileptic patients and patients at risk of convulsions.
- The oral solution is not recommended in children aged less than 2 years.
- Concurrent use with alcohol or other CNS depressants may cause additional reductions in alertness and impairment of performance.
- Caution in elderly patients, patients with hepatic dysfunction and renal insufficiency is recommended.

Effects on the ability to drive and use machines

The occurrence of somnolence has been reported in some patients taking cetirizine. Hence, caution should be exercised when driving a car or operating potentially dangerous machinery.

PREGNANCY AND LACTATION

Pregnancy

Caution should be exercised when prescribing to pregnant women.

Lactation

Caution should be exercised when prescribing cetirizine to lactating women.

Cetirizine is excreted in human milk at concentrations representing 25% to 90% of those measured in plasma depending on sampling time after administration.

DRUG INTERACTIONS

Due to pharmacokinetic, pharmacodynamic and tolerance profile of cetirizine, no interactions are expected with this antihistamine. Actually, neither pharmacodynamic nor significant pharmacokinetic interaction was reported in drug-drug interactions studies performed, notably with pseudoephedrine or theophylline (400 mg/day).

The extent of absorption of cetirizine is not reduced with food, although the rate of absorption is decreased.

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MAIN SIDE/ADVERSE EFFECTS

- **Cardiac disorders**
Tachycardia (rare)
- **Blood and lymphatic Disorders**
Thrombocytopenia (Very rare)
- **Eye disorders**
Accommodation disorder, blurred vision, oculogyration (Very rare)
- **Gastro-intestinal disorders**
Diarrhea (Uncommon), xerostomia
- **Renal and urinary disorders**
Dysuria, enuresis (Very rare)
- **Skin and subcutaneous tissue disorders**
Pruritus, rash (Uncommon), urticaria (Rare), angioneurotic oedema, fixed drug eruption (Very rare)
- **General disorders and administration site conditions**
Asthenia, malaise (Uncommon), oedema (Rare), fatigue
- **Immune system disorders**
Hypersensitivity (Rare), anaphylactic shock (Very rare)
- **Hepatobiliary disorders**
Hepatic function abnormal (increased transaminases, alkaline phosphatase, γ -GT and bilirubin) (Rare)
- **Psychiatric disorders**
Agitation (Uncommon), aggression, confusion, hallucination, insomnia (Rare), tic (Very rare)
- **Endocrine/Metabolic Effects**
Weight gain (Rare)
- **Neurologic Effects**
Asthenia, somnolence, headache, sedation, paraesthesia (Uncommon), convulsions (Rare), dysgeusia, syncope, tremor (Very rare), dyskinesia, dystonia (Very rare), dizziness

OVERDOSE

Symptoms

Symptoms observed after an overdose of Cetirizine are mainly associated with CNS effects that could suggest an anticholinergic effect. Adverse effects reported after an intake of at least 5 times the recommended daily dose are: confusion, diarrhea, dizziness, fatigue, headache, malaise, mydriasis, pruritus, somnolence, tachycardia, restlessness, sedation, tremor and urinary retention.

Treatment

There is no specific antidote for Cetirizine. Should overdose occur, symptomatic or supportive treatment is recommended. Gastric lavage should be considered following ingestion of a short occurrence. Cetirizine is not effectively removed by dialysis.

DOSAGE AND ADMINISTRATION

Oral. May be given without regard to meals.

Children:

- *Children aged from 2 to 6 years:*
2.5 mg (2.5 ml of oral solution) twice daily.
- *Children aged from 6 to 12 years:*
5 mg (5 ml of oral solution) twice daily.
- *Children over 12 years of age:*
10 mg (10 ml of oral solution) once daily.

Adults:

10 mg once daily (10 ml oral solution). A 5 mg starting dose (5 ml oral solution) may be proposed if this leads to satisfactory control of the symptoms. The solution can be swallowed as such.

Elderly subjects:

Data do not suggest that the dose needs to be reduced in elderly subjects provided that the renal function is normal.

Patients with moderate to severe renal impairment:

The dosing intervals must be individualized according to renal function. Refer to the following table and adjust the dose as indicated. To use this dosing table, an estimate of the patient's creatinine clearance (CL_{cr}) in ml/min is needed.

Dosing Adjustments for Adult Patients with Impaired Renal Function Group:

Dosing Adjustments for Adult Patients with Impaired Renal Function Group	Creatinine clearance (ml/min)	Dosage and frequency
Normal	≥ 80	10 mg once daily
Mild	50 – 79	10 mg once daily
Moderate	30 – 49	5 mg once daily
Severe	< 30	5 mg once every 2 days
End-stage renal disease - patients undergoing dialysis	< 10	Contra-indicated

In paediatric patients suffering from renal impairment, the dose will have to be adjusted on an individual basis taking into account the renal clearance of the patient and his body weight.

Patients with hepatic impairment:

No dose adjustment is needed in patients with solely hepatic impairment.

Patients with hepatic impairment and renal impairment:

Adjustment of the dose is recommended.

Storage:

Store below 30°C. Protect from light.

Presentation/ packing:

Syrup 5 mg / 5 mL x 30 mL, 60 mL and 120 mL.

Product Registration Holder /

Manufactured by: HOVID Bhd.
121, Jalan Tunku Abdul Rahman
(Jalan Kuala Kangsar),
30010 Ipoh, Perak, Malaysia.

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